

face detection and face expression recognition. You should end up with the file/folder structure shown in Figure 1.

Testing if everything is installed properly

Now open the *index.html* file and add the following code to it:

```
<!doctype html>

<head>
  <title>Face Expression Recognition</title>
  <script type="text/javascript"
src="node_modules/face-api.js/dist/face-api.min.js"></script>
</head>

<body>
  <h1>Hello World</h1>
</body>

</html>
```

Finally, start the server by running the following command at the prompt:

```
http-server ./
```

This command will start the HTTP server. Now, load *http://127.0.0.1:8080/index.html* in your browser to run the project. If you see the output screen shown in Figure 2, then you have successfully installed all the dependencies required.

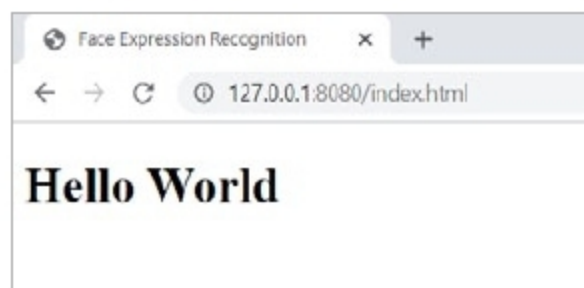


Figure 2: The project running in the browser

Creating the app

Now, let's create a facial expression recognition app, which will detect the faces and expressions of the user. We are going to use pre-trained models to detect the face and recognise expressions. You

can download these from *http://www.aniket.co/labs/face/models.zip*. We will use these models in the app. You need to extract these into the 'Models' folder inside your project directory.

Note: A webcam is needed for this project, as we are detecting the face live from video. So it is recommended that you try this project on a laptop with a webcam.

Now, open *index.html* and copy the following HTML code:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-scale=1.0">
  <meta http-equiv="X-UA-Compatible"
content="ie=edge">
  <title>Face Expression Recognition</title>
  <script defer src="node_modules/face-api.js/dist/face-api.min.js"></script>
  <script defer src="script.js"></script>
  <style>
    body {
      margin: 0;
      padding: 0;
      width: 100vw;
      height: 100vh;
      display: flex;
      justify-content: centre
      align-items: centre
    }

    canvas {
      position: absolute;
    }
  </style>
</head>

<body>
  <video id="video" width="720"
height="560" autoplay muted></video>
```

```
</body>
</html>
```

As you can see in the HTML code above, we have used the *<script>* tag to import *face-api* and *script.js* files. In the *<style>* tag we have added some styling to the elements; basically, we have centred the UI. And finally, we have used the *<video>* tag, which will display that a face has been detected and recognised via webcam.

Note: We have added auto-play and muted the attribute to the video tag (since we don't need audio) with this webcam, which will start capturing videos as soon as the app is launched.

Now open *script.js* and add the following JavaScript code:

```
const video = document.
getElementById('video')

Promise.all([
  faceapi.nets.tinyFaceDetector.
loadFromUri('/models'),
  faceapi.nets.faceLandmark68Net.
loadFromUri('/models'),
  faceapi.nets.faceRecognitionNet.
loadFromUri('/models'),
  faceapi.nets.faceExpressionNet.
loadFromUri('/models')
]).then(startVideo)

function startVideo() {
  navigator.getUserMedia({ video: {}
},
  stream => video.srcObject =
stream,
  err => console.error(err)
)

video.addEventListener('play', () => {
  const canvas = faceapi.
createCanvasFromMedia(video)
  document.body.append(canvas)
  const displaySize = { width: video.
width, height: video.height }
```